🎨 CampusPulse — Frontend Architecture & UI Engineering Manual

Version: 1.0.0 (Production)

Classification: Frontend Engineering Specification

Framework: Next.js 16.3.3 (App Router) + React 19.2.8 + Tailwind CSS v4

Last Updated: September 2026

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1. Frontend Architecture Overview

CampusPulse is engineered around the **Next.js 16 App Router paradigm**, leveraging React 19 Server Components for data-heavy views and Client Components (`"use client"`) for interactive elements. This architecture minimizes the JavaScript bundle shipped to the client, guaranteeing near-instant first contentful paint (FCP) and optimal SEO scores.

Key Architectural Pillars:

1. **Server-Driven Data Fetching:** Initial event catalogs, admin dashboards, and legal documents are streamed directly from server components without client-side `useEffect` waterwalls.
2. **Zero-Flicker Authentication:** NextAuth `SessionProvider` wrapped in `src/components/providers/AuthProvider.tsx` maintains persistent client session states.
3. **Glassmorphic Design System:** Custom Tailwind CSS v4 design tokens built for low-glare dark mode (`#08090D` background, `#12151E` surface cards, `#6366F1` indigo highlights).
4. **Fluid Motion & Accessibility:** Page transitions and interactive drawer modals powered by Framer Motion, compliant with WCAG 2.1 AA standards.

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2. Complete Routing Directory Map (All 32 Routes)

Below is the exhaustive index of all 32 compiled routes in CampusPulse:

Route Path	Type	Access Level	Description
`/`	Static	Public	CampusPulse landing page, hero section, live stats counter, upcoming events showcase
`/events`	Static	Public	Searchable, category-filtered campus event directory
`/events/[slug]`	Dynamic	Public / Auth	Full event details, live seat progress bar, schedule, RSVP trigger modal
`/events/verify/[token]`	Dynamic	Public	Official Gate Pass Scanner page for security guards & event coordinators
`/portal/qna`	Dynamic	Auth (Student/Admin)	Live event Q&A board, question upvoting, official replies
`/portal/feedback`	Dynamic	Auth (Student/Admin)	Active student feedback surveys catalog
`/portal/feedback/[formId]`	Dynamic	Auth (Student/Admin)	Dynamic survey form renderer with star ratings and comments
`/portal/billing`	Dynamic	Auth (Student/Admin)	Transaction summary and digital event pass receipt overview
`/portal/billing/success`	Dynamic	Auth (Student/Admin)	Successful ticket confirmation with downloadable PDF receipt
`/portal/billing/pending`	Dynamic	Auth (Student/Admin)	Real-time payment verification waiting screen
`/portal/billing/failed`	Dynamic	Auth (Student/Admin)	Error diagnosis and retry transaction launcher
`/admin`	Dynamic	Admin Only	Primary institutional administration KPI dashboard
`/admin/events`	Dynamic	Admin Only	Event management table with status toggles (Draft/Upcoming/Completed)

Route Path	Type	Access Level	Description
`/admin/events/new`	Dynamic	Admin Only	Comprehensive event creation wizard (Dates, venue, capacity, tags)
`/admin/events/[id]/edit`	Dynamic	Admin Only	Event modification and live attendance editor
`/admin/feedback`	Dynamic	Admin Only	Google-Forms-style survey builder with dynamic fields
`/admin/feedback/[eventId]`	Dynamic	Admin Only	Feedback configuration for specific campus events
`/admin/feedback/[eventId]/responses`	Dynamic	Admin Only	Visual sentiment charts and student response data tables
`/admin/media`	Dynamic	Admin Only	High-resolution photo vault & MP4/WebM video gallery manager
`/admin/analytics`	Dynamic	Admin Only	Recharts-powered graphs: RSVP trends, category popularity, sentiment
`/admin/reports`	Dynamic	Admin Only	Executive report generator with printable letterhead and PDF download
`/admin/settings`	Dynamic	Admin Only	Campus brand settings, hero banner updates, social broadcast links
`/login`	Static	Public	Secure credential sign-in page with role routing
`/register`	Static	Public	Student onboarding registration with department selector
`/forgot-password`	Static	Public	Self-service password recovery email dispatcher
`/reset-password`	Static	Public	Encrypted token-based credential reset interface
`/onboarding`	Static	Public	Interactive campus tour explaining platform features
`/about`	Static	Public	Mission statement, university committee leadership, and

Route Path	Type	Access Level	Description
			contacts
`/help`	Static	Public	Searchable student FAQ, troubleshooting, and support desk
`/legal`	Static	Public	University compliance, privacy, terms, and WCAG accessibility index
`/legal/[slug]`	Dynamic	Public	Specific policy view (Privacy Policy, Terms of Service, Cookie Policy)
`/sitemap.xml`	Dynamic	SEO	Auto-generated XML sitemap covering all static & live event URLs
`/robots.txt`	Static	SEO	Search engine crawler rules (protects `/admin` & `/portal/billing`)

3. UI Component Hierarchy & Design Tokens

CampusPulse avoids generic component libraries in favor of tailored, highly performant React components:

```
src/components/
├── layout/
│   ├── Navbar.tsx            # Sticky frosted-glass navbar with role-aware navigation
│   └── Footer.tsx            # Multi-column university footer with quick links & badges
├── providers/
│   ├── AuthProvider.tsx      # NextAuth SessionProvider wrapper
│   └── ThemeProvider.tsx     # Dark glassmorphic theme state controller
├── ui/
│   ├── CookieConsentBanner.tsx # GDPR/ePrivacy compliant banner with storage preferences
│   ├── Button.tsx            # High-contrast interactive button with loading spinner
│   ├── Modal.tsx             # Accessible backdrop-blurred modal dialog
│   └── Badge.tsx             # Category color tags (Technical, Cultural, Sports, Workshop)
└── media/
    ├── VideoPlayer.tsx       # Custom HTML5 video player with buffer indicators
    └── ImageLightbox.tsx     # Full-screen photo inspector with zoom & share
```

Color Palette Tokens:

- **Canvas Base:** `#08090D` (Deep obsidian black for zero eye fatigue)
- **Surface Elevation 1:** `#111420` (Cards, modals, navigation drawers)

- **Surface Elevation 2:** `#1A1F30` (Hover states, table headers, input fields)
- **Primary Accent:** `#6366F1` (Indigo electric highlight for primary CTAs)
- **Secondary Accent:** `#8B5CF6` (Purple gradient complement)
- **Success Emerald:** `#10B981` (Confirmed RSVPs, verified pass status)
- **Warning Amber:** `#F59E0B` (Seats running low, pending payments)
- **Danger Crimson:** `#EF4444` (Event cancellations, rejected transactions)

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4. Client State & Form Validation

- **Form Handling:** Forms leverage controlled inputs or native `FormData` dispatched via asynchronous Server Actions.
- **Validation:** Client-side feedback is powered by **Zod schemas**, preventing invalid emails, weak passwords, and mismatched date ranges before network dispatch.
- **Visual Feedback:** All asynchronous operations show micro-spinners, disabled state locks, and animated alert banners upon resolution.
